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Amount of clockwise rotation of Southwest Japan —fan shape opening of the southwestern part of the Japan Sea

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Paleomagnetic measurements have been carried out on welded tuffs ranging in age between 58 Ma and 112 Ma from Yamaguchi and Gō river areas in the central part of Southwest Japan. The new data, together with those of younger igneous rocks published previously, define the change of paleomagnetic field direction during the late Mesozoic/Cenozoic period for Southwest Japan. The paleomagnetic direction from this area has pointed 56 ± 3° clockwise from the expected field direction estimated from APWP (apparent polar wandering path) of the whole of Eurasia during the period between 100 and 20 Ma. In comparison with the expected one from the eastern margin of Eurasia (Korea, China, Siberia), the Cretaceous field direction of Southwest Japan shows the clockwise deflection by 44–49°. These results establish that while the eastern margin of Eurasia, including Southwest Japan, was rotated more or less with respect to the main part of Eurasia during last 100 Ma, Southwest Japan was rotated clockwise through more than 40° with respect to the eastern margin of Eurasia since 20 Ma. The large amount of rotation for Southwest Japan implies that it is rotated by an opening of the southwestern part of the Japan Sea, which widens northeastward (fan shape opening). The tectonic feature of Southwest Japan and the Japan Sea is analogous to that of Corso-Sardinia and the Ligurian Sea in the Mediterranean, indicating that the fan shape opening is a specific feature of the rifting of the continental sliver at the continental rim.

1. Introduction

The Japan arc has been recognized to be a continental sliver rifted from the Eurasian continental margin following the early speculations by Terada [1] and Carey [2]. Several forms of geological and geophysical evidence from the Japan Sea supports this rifting hypothesis of the Japan arc [3–8]: the Japan Sea has originated from sea-floor spreading, which consequently caused the Japan arc to drift away from the continent.

The Japan arc is a rare example of the rifting process of continental slivers, although such rifting of continental slivers would be expected to have occurred in many places since the breakup of Gondwana [9]. Because the rifting of the Japan arc is presumed to have occurred as recently as the middle Miocene [5,10,11], the geomorphological and geological evidence has not been obscured by subsequent geological activity. The Japan arc thus provides an opportunity to study the geodynamics of the rifting of the continental sliver.

One of the most important features of the rifting process is the mode of opening of a basin in association with rifting of the sliver from its mother continent, because the rifting mode sheds light on the understanding for the origin of the forces affecting the continental margin and for the mechanical resistance of the continental lithosphere. Some models have been proposed for the rifting of the continental sliver; (1) fan shape opening mode of the back arc basin [12], (2) parallel opening mode of the Atlantic type [13], and (3) pull-apart basin [14,15]. Drifting motion of a continental sliver is the clue to inference for the opening mode, because the difference between the fan shape mode and the other two modes can be distinguished by the degree of rotation during drift of the sliver.

Paleomagnetism offers a practicable method for estimating the amount of rotation of the continental sliver. The aim of this paper is to discuss the rifting model for the Southwest Japan–Japan Sea system on the basis of the paleomagnetic results.

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An apparent polar wandering path (APWP) has been obtained for Southwest Japan for the period since 100 Ma, and it is compared with that from Eurasia.

2. Geology and sampling

Our apparent polar wandering path (APWP) has been obtained for the Yamaguchi and Gō river areas (extent of 150 km × 50 km) in the central part of Southwest Japan (See Fig. 1). The areal extent of the coherent tectonic unit of Southwest Japan is now disputed [16], and consequently paleomagnetic data compiled from the entire area of Southwest Japan, as defined geographically, provide unwanted information about the different block motions from regions marginal to the coherent block. Our study area is limited to the central part of this area, so that our data are appropriate for obtaining the characteristic APWP for the Southwest Japan block.

Extensive volcanic activity accompanied with intrusion of plutonic rocks occurred in the Inner Zone of Southwest Japan between the late Mesozoic and Tertiary periods (Fig. 1). The volcanic rocks are mostly welded pyroclastics with interbedded lavas and clastic rocks. The Yamaguchi to Gō river areas have been geologically mapped in detail, and extensive radiometric dating has been done using the K-Ar, Rb-Sr and fission-track methods [17–20]. The late Mesozoic to Tertiary succession has been established as shown in Fig. 1 [17], so that the APWP is traced back to 100 Ma.

In Yamaguchi area, Cretaceous andesitic to rhyolitic volcanism formed three groups called the Kanmon, Shunan and Abu. The volcanism occurred during the same or during a slightly younger period at the Gō river area forming the Ochi and Sakurae Groups which are composed of rhyolitic to dacitic pyroclastics and rhyolitic lava. In the Oligocene to the Miocene, continued dacitic to andesitic volcanism, associated with cauldron subsidence in most cases, produced a younger volcanic unit (the Kawauchi and Iwami Groups) in the Gō river and neighbouring regions.

Our paleomagnetic approach is to measure the stable natural remanent magnetization (NRM) of the thermal origin of igneous rocks aged between 50 and 100 Ma. More than 400 samples were collected from the Shunan, Abu, Ochi and Sakurae Groups at 41 sites (see Fig. 1), listed in Table 1.

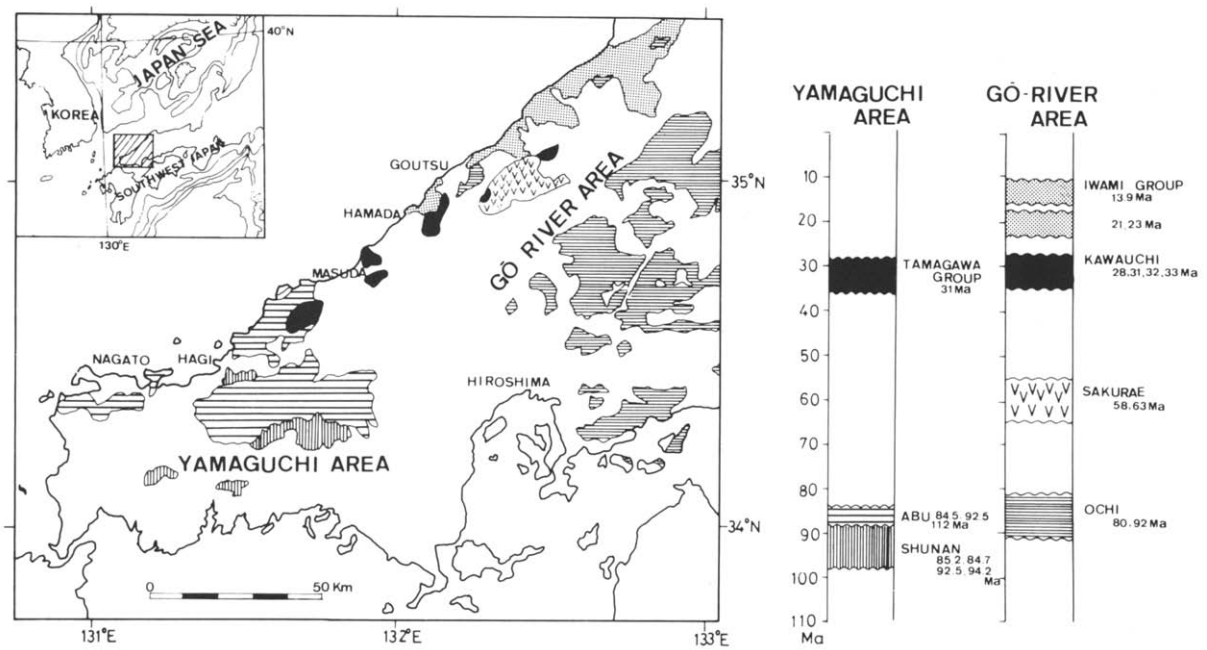


Fig. 1. Distribution of late Mesozoic/Cenozoic volcanic rocks and their stratigraphic relation in Yamaguchi and Gō river areas in Southwest Japan. The ages of radiometric dating [17–20] are assigned to each group.

Detailed paleomagnetic studies have been already reported for igneous rocks younger than 40 Ma in this area [10].

All the samples were collected from welded tuff, because it possesses stable remanent magnetization [10,11,21–23] and any tilting could be corrected on the basis of the eutaxitic structure (lineation of stretched pumice and aligned phenocrysts) [24]. Sampling was carried out only at outcrops where the eutaxitic structure was clearly observed. The mean of tilting dips in the outcrops was 17°. Since the eutaxitic structure does not always indicate the paleohorizontal, our sampling sites were selected over as wide an area as possible in order to maximize the chance that the original dips have averaged to zero.

Hand samples were taken and a magnetic compass was used for orientation. Each site typically comprised twelve independently oriented samples distributed over distances ranging up to 20 m.

3. Paleomagnetism

In the laboratory, individual specimens, 25 mm in diameter and 25 mm long, were prepared from the samples. Natural remanent magnetizations (NRMs) were measured with either a Schonstedt SSM-1A spinner magnetometer or a ScT cryogenic magnetometer according to the intensity of magnetization. The stability of remanence was investigated by both thermal and alternating field (AF) methods. Specimens were thermally demagnetized using a laboratory built furnace. The residual magnetic field in which the specimens were cooled was less than 10 nT. AF demagnetization was carried out with three-axis tumbler in one cylindrical mu-metal shield. Zijderveld component plots [25] were used to assess the directional stability and coercivity spectrum of each specimen.

Three pilot specimens from each site were thermally demagnetized at more than seven steps up to the Curie temperature. One pilot specimen was progressively AF demagnetized up to 50 mT.

The results of thermal demagnetization from 38 of the 41 sites agree very well with the AF demagnetization results (Fig. 2A, B). For 30 of the 38 sites, the magnetization vectors decay more or less uniformly towards the origin: the change in NRM directions was less than 10° (Fig. 2C, D). Some specimens from the other eight sites possessed secondary component of magnetization. There was a great shift of more than 60° in the NRM directions of the specimens from both IM704 and IM709 during demagnetization runs (Fig. 2E). This component was usually removed at either 500°C or 10–30 mT. This is most likely a viscous remanent magnetization (VRM) induced by the recent geomagnetic field. The thermal demagnetization method appeared to be effective for removing the secondary component.

One specimen from all independently oriented samples for 36 of the 38 sites was thermally demagnetized at an optimal temperature. The demagnetization temperature was chosen to produce minimum dispersion for NRM directions from three pilot specimens. All the specimens for two sites (IM704 and IM709) with strong secondary component were progressively demagnetized. The stable end points during demagnetization runs were identified to be their characteristic directions. The demagnetization vectors of the 38 sites reached the origin at around 580°C, indicating that the remanent magnetization resided in magnetite rather than hematite.

Stable behaviour was observed in the specimens from two sites (IM614 and IM627) of welded tuffs of the Shunan Group during stepwise AF demagnetization runs up to 50 mT and during thermal demagnetization up to 400°C. Direction became random on further thermal demagnetization. Demagnetization at 400°C produced less than a 50% drop in the intensity magnetization, indicating that the thermoremanent magnetizations (TRMs) were probably acquired during the temperature interval between 300° and 400°C. We accepted the NRM after thermal cleaning at 300°C and 400°C as the characteristic remanent magnetization for IM614 and IM627, respectively.

Pilot specimens for only one site from the Shunan Group failed to converge in a stable direction during stepwise AF and thermal demagnetization runs. No further work was done on this site.

In order to exclude poorly determined or spurious directions from the data set, we arbitrarily discarded results from any site for which the circles of 95% confidence exceeded 15°. Only two sites failed this criterion. The paleomagnetic data before and after tilt correction for the acceptable sites are listed in Table 1.

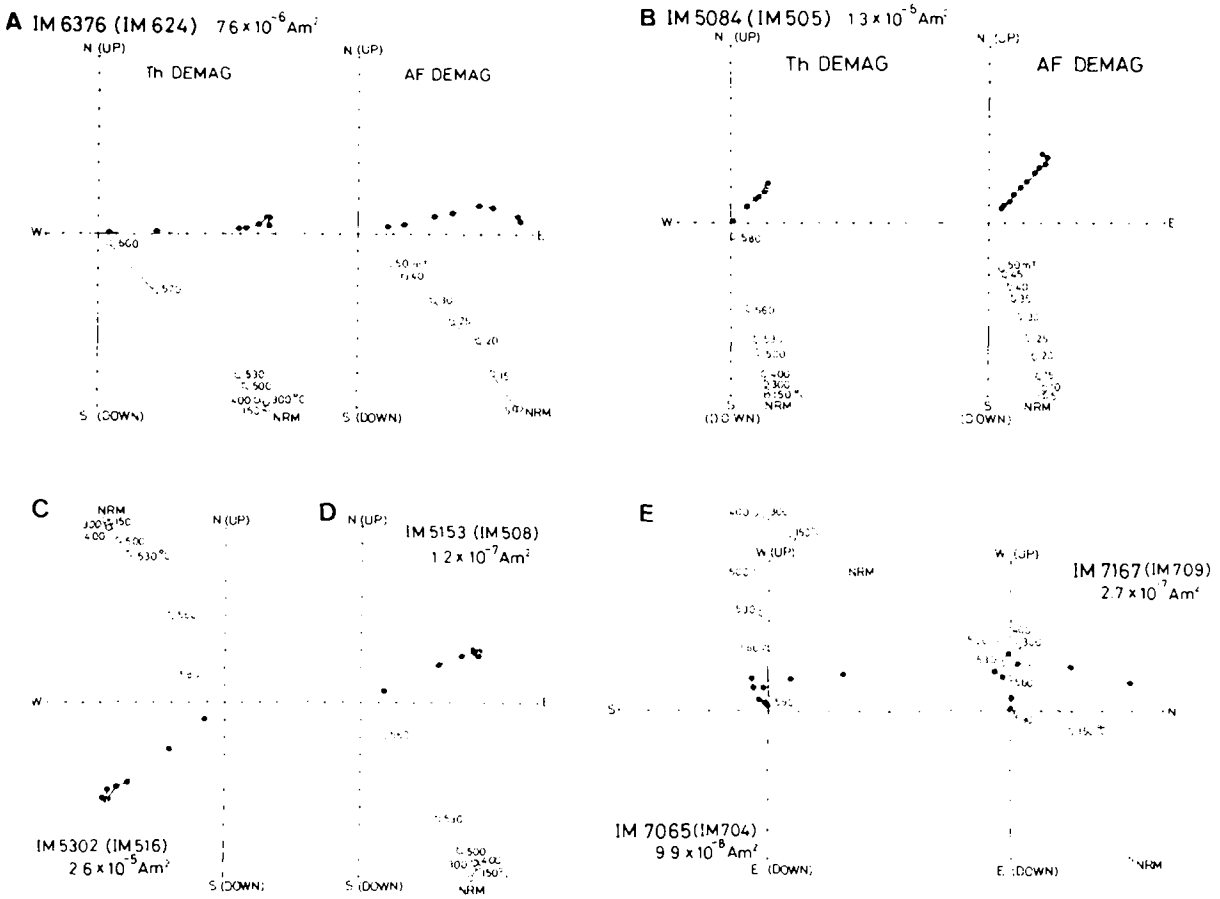


Fig. 2. Orthogonal projection plots for (A) twin specimens from the Abu Group (the Ebune Formation) (Th DEMAG. = thermal demagnetization, AF DEMAG. = alternating field demagnetization); (B) twin specimens from the Abu Group (the Oumi Formation); (C) a specimen with reversed polarity from the Sakurae Group (the Nakano Formation N4); (D) a specimen with normal polarity from the Ochi Group (the Yadani Formation); and (E) two specimens with multi-component magnetization from the Sakurae Formation (the Nakano Formation N4). Open (solid) symbols show the magnetic vectors projection on to the vertical (horizontal) plane. Tectonic tilt corrections have not been applied.

The paleomagnetic data are divided into three sets on the basis of their radiometric ages; 112–85 Ma, 92–80 Ma and 63–58 Ma. Site mean directions were combined with published data [10,23] to obtain the mean direction of each age interval (Fig. 3 and Table 2). The paleomagnetic data for the youngest interval [10,23] are also shown in Fig. 3 and Table 2. The mean paleomagnetic direction in Table 2 is presented as the value at a representative point (134°E, 35°N) of Southwest Japan.

3.1. 85–112 Ma (the Shunan and Abu Groups)

Samples from all the sites have normal polarity. They have consistently clockwise-deflected magnetic declination from 9° to 116°.

The paleomagnetic directions after tilt correction are tightly clustered. Correcting for dip, however, did not reduce the scatter of the directions ($D_i = 55.4°$, $I_i = 65.6°$, $k = 55.7$, $\alpha_{95} = 4.3°$ vs. $D_c = 64.6°$, $I_c = 63.7°$, $k = 30.7$, $\alpha_{95} = 5.8°$). Difference in radius of 95% confidence is only 1.5°. Uncertainty in determinating the dip for welded tuff is fairly large: While the error in dip measurement is no greater than 5°, the error in stike measurement is of the order of ±5°. Errors are further introduced in tilt correction by the usual way of "unfolding", that is, rotation of the bed-

TABLE 1

Summary of paleomagnetic results for the Yamaguchi and Gō river areas

Geological sequence	Age	Site	Demagnetization	N	Tilt correction		In situ		α_{95}(°)	k	Locality		Rock type
					D (°)	I (°)	D (°)	I (°)			long. E	lat. N	
Sakurae Group													
Nakano Formation	N5	IM517	400 A	11	−118.8	−60.8	(−156.7	−56.0)	2.7	296.1	132°27.3′	34°56.8′	D WT
	N4	IM516	400 A	9	−131.2	−55.9	(−152.3	−48.0)	4.0	170.0	132°27.2′	34°56.9′	D WT
		IM704	560 B	11	−144.0	−73.4	(−149.0	−68.0)	8.1	32.5	132°28.1′	34°57.4′	D WT
		IM708	500 A	12	−117.9	−55.5	(−143.7	−57.2)	2.9	229.4	132°25.8′	34°56.0′	D WT
		IM709	530 B	11	−99.1	−54.8	(−118.2	−56.6)	8.9	27.1	132°24.7′	34°56.0′	D WT
		IM216 *	500 A	9	−115.8	−56.0	(−142.1	−50.5)	18.9	8.4	132°27.7′	34°57.5′	D WT
	N3 63 Ma (F.T.)	IM515	400 A	11	−92.4	−78.2	(−147.8	−57.40	3.3	189.5	132°26.8′	34°57.3′	D WT
		IM106‡	500 A	12	−100.2	−27.0	(−113.3	−53.2)	8.9	22.5	132°28.7′	34°59.0′	D WT
		IM217 ‡	400 A	11	−118.4	−64.0	(−154.6	−58.5)	2.7	289.5	132°26.4′	34°57.2′	D WT
	N2 58 Ma (F.T.)	IM706	400 A	10	74.6	41.8	(76.4	22.0)	3.6	179.7	132°25.8′	34°57.1′	D WT
		IM707	530 A	12	70.5	23.8	(66.6	11.7)	7.9	31.4	132°25.8′	34°57.0′	D WT
		IM108 ‡	500 B	8	−107.7	−15.8	(−118.2	−23.0)	16.3	12.5	132°26.5′	34°57.4′	D WT
		IM109 ‡	400 B	10	−64.4	−72.9	(−123.8	−79.5)	2.7	325.3	132°25.7′	34°57.6′	D WT
		IM218 ‡	500 A	11	98.8	10.9	(95.4	19.2)	3.4	181.1	132°26.1′	34°57.2′	D WT
Ochi Group													
Yadani Formation	80 ≪ 92 Ma (F.T.)	IM506	500 B	11	−81.2	0.5	(−81.1	−47.5)	8.2	31.7	132°14.7′	35°00.0′	RD WT
		IM507	400 B	12	−116.3	−31.3	(−103.2	−36.0)	6.8	41.2	132°33.6′	34°57.9′	RD WT
		IM508	300 A	11	52.0	55.9	(71.8	60.8)	4.1	125.0	132°31.1′	34°55.8′	RD WT
		IM509	300 A	10	86.5	44.2	(71.7	44.4)	4.4	122.7	132°31.4′	34°55.5′	RD WT
		IM510	300 A	10	73.7	55.5	(84.2	60.1)	7.9	38.0	132°31.7′	34°54.9′	RD WT
		IM511	400 B	11	77.3	47.9	(68.5	50.6)	2.6	301.0	132°31.4′	34°53.0′	RD WT
		IM111 ‡	500 B	11	−131.5	−44.7	(−161.0	−24.8)	6.7	47.1	132°28.5′	34°56.5′	RD WT
		IM111′ ‡	500 B	6	−91.1	−36.4	(−131.2	−45.2)	13.5	25.6	132°28.5′	34°56.5′	MS
		IM205 ‡	500 A	12	−119.7	−63.1	(−95.8	−52.6)	2.2	377.2	132°33.1′	34°58.0′	RD WT
Ichiibara Formation	92 Ma (F.T.)	IM512	400 B	12	−101.6	−43.1	(−116.1	−49.0)	4.6	90.1	132°28.5′	34°56.9′	R WT
		IM513	400 B	11	−121.1	−31.9	(−144.7	−27.1)	4.9	88.4	132°29.0′	34°56.9′	R WT
		IM514	400 A	9	−116.0	−37.3	(−112.0	−29.4)	6.0	75.1	132°29.6′	34°56.9′	R WT
		IM112 ‡	500 A	8	−72.7	−53.4	(−119.5	−59.4)	4.9	130.6	132°28.3′	34°56.7′	R WT
		IM206 ‡	500 A	11	−137.4	−35.8	(−146.0	−54.0)	6.4	51.5	132°30.7′	34°58.3′	R WT
		IM1137 ‡	550 B	9	−99.3	−18.2	(−109.7	−19.3)	9.7	29.0	132°30.8′	34°58.8′	R WT
Abu Group	92.1, 84.5 Ma (F.T.); 112 Ma (Rb-Sr)												
Ebune Formation		IM501	300 A	11	59.4	53.6	(59.4	53.6)	2.8	270.0	131°41.4′	34°24.2′	R WT
		IM505	300 A	11	40.9	71.8	(45.7	60.0)	3.3	187.9	131°32.6′	34°21.9′	RD WT
		IM618	400 A	10	67.2	46.9	(75.7	49.3)	1.7	829.7	131°30.6′	34°19.2′	D WT
		IM619	400 A	11	70.1	43.2	(77.7	54.5)	2.6	312.0	132°30.6′	34°18.8′	D WT
		IM620	400 A	11	48.2	66.1	(48.2	66.1)	3.0	232.5	131°28.8′	34°18.8′	R WT
		IM621	400 A	11	89.8	70.6	(47.4	65.9)	2.7	294.3	131°26.0′	34°18.8′	R WT
		IM622	400 A	10	65.9	51.9	(44.1	67.9)	2.3	445.6	131°24.6′	34°18.8′	R WT
Oumi Formation		IM623	400 A	12	66.8	66.4	(45.4	48.6)	3.7	137.3	131°10.6′	34°24.2′	R WT
		IM624	400 A	12	77.4	44.6	(46.4	68.1)	3.6	142.8	131°13.4′	34°25.4′	D WT
		IM625	400 A	10	116.5	60.0	(69.1	70.5)	3.1	244.3	131°13.4′	34°25.5′	D WT

TABLE 1 (continued)

Geological sequence	Age	Site	Demagnetization	N	Tilt correction		In situ		α_{95}(°)	k	Locality		Rock type
					D (°)	I (°)	D (°)	I (°)			long. E	lat. N	
Maitani Formation		IM504	300 A	12	75.6	54.1	(45.5	63.6)	2.6	289.4	131°33.7′	34°19.7′	R WT
		IM615	400 A	12	39.8	62.7	(24.5	70.3)	3.0	216.5	131°28.6′	34°15.0′	R WT
		IM616	400 A	12	56.7	53.7	(60.7	66.5)	2.8	242.6	131°30.5′	34°17.8′	R WT
		IM617	400 A	11	50.8	74.0	(60.2	54.4)	5.0	85.2	131°30.4′	34°18.3′	R WT
Shunan Group	92.5, 85.2, 94.2 Ma (F.T.); 84.7 Ma (K-Ar)												
Monomidake Formation		IM503	400 A	12	71.4	82.8	(91.2	63.4)	2.0	465.9	131°38.4′	34°17.0′	R WT
		IM601	400 A	11	101.1	53.4	(68.9	54.0)	2.1	488.5	131°29.3′	34°07.4′	R WT
		IM611	400 A	11	32.9	55.2	(43.6	62.9)	3.9	140.8	131°49.7′	34°14.4′	RD WT
		IM612	400 A	10	66.3	67.7	(36.2	57.7)	4.7	104.6	131°46.9′	34°14.8′	R WT
		IM614	300 B	9	47.6	52.9	(39.1	75.6)	8.2	32.3	131°46.3′	34°13.9′	R WT
		IM626	400 A	10	57.9	62.3	(26.8	80.1)	3.6	176.5	131°42.8′	34°17.0′	R WT
		IM627	400 A	12	9.4	66.8	(44.9	71.0)	13.4	11.4	131°41.8′	34°17.4′	RD WT

Both paleomagnetic directions before (in situ) and after tilt correction (tilt corrected) are presented. Data denoted by asterisk are from Otofuji and Matsuda [10,23]. Demagnetization: the magnetic cleaning temperature and behaviour of NRM direction during thermal demagnetization; A = single-component magnetization, change in NRM direction was less than 10° during demagnetization. B = multi-component magnetization, change was more than 10°. N is the number of samples in site mean computation. D is the site mean declination, I is the site mean inclination, α_{95} is the radius of the 95% circle of confidence around the site mean direction and k is Fisher precision parameter. Rock type: D WT = dacite welded tuff, R WT = rhyolite welded tuff, RD WT = ryhodacite welded tuff; and MS = mud stone. The ages are data after Murakami [17], Shibata et al. [18], Kano and Yoshida [19] and Matsuda [20]. F.T = Fission track dating.

ding plane into the horizontal around a horizontal axis parallel to the strike. It seems likely that the errors in making correction for tectonic deformation increases the scatter for directions after tilt correction.

The following evidence implies that welded tuffs of the Shunan and Abu groups have stable remanence associated with their formation:

(1) The stratigraphically concordant ages between 85 and 94 Ma have been obtained by the

TABLE 2

Average paleomagnetic directions and poles for Southwest Japan

Stage (Ma)	Age interval (Ma)	N	D (°)	I (°)	α_{95} (°)	ΔD (°)	VGP		α_{95} (°)	R (°)	ΔR (°)
							long.	lat.			
13.9		4	−8.5	56.0	17.5	32.5	68.4° E	82.7° N	21.1	−8.5	32.5
20	20– 30	12	60.0	41.8	10.9	14.7	143.1° W	37.9° N	10.9	54.4	15.6
			(73.2	32.4	12.2)						
30	30– 35	26	65.9	48.8	6.5	9.9	151.5° W	36.0° N	7.4	56.2	11.9
			(70.5	44.1	8.8)						
60	58– 63	14	74.1	52.1	12.0	19.8	161.0° W	32.4° N	12.4	60.1	21.6
			(54.5	50.9	12.6)						
80	80– 92	14	73.4	43.0	10.0	13.7	150.0° W	28.0° N	9.8	57.8	16.8
			(63.0	47.4	9.8)						
100	85–112	21	64.6	63.7	5.8	13.2	170.6° W	41.5° N	8.5	52.7	14.8
			(55.4	65.6	4.3)						

Directions are presented as the value at the representative point of Southwest Japan (134° E, 35° N). Both directions before and after tilt correction: Directions before tilt correction is shown in round brackets ( ). Data denoted by asterisk are after Otofuji and Matsuda [10,23]. N is the number of sites, D, I are mean declination and inclination, α_{95} and k are Fisher statistic parameter. R is the difference between declination value of Southwest Japan and that at the reference point expected from VGP for Eurasia. ΔD and ΔR are uncertainty in declination and rotation at 95% confidence, respectively (see text).

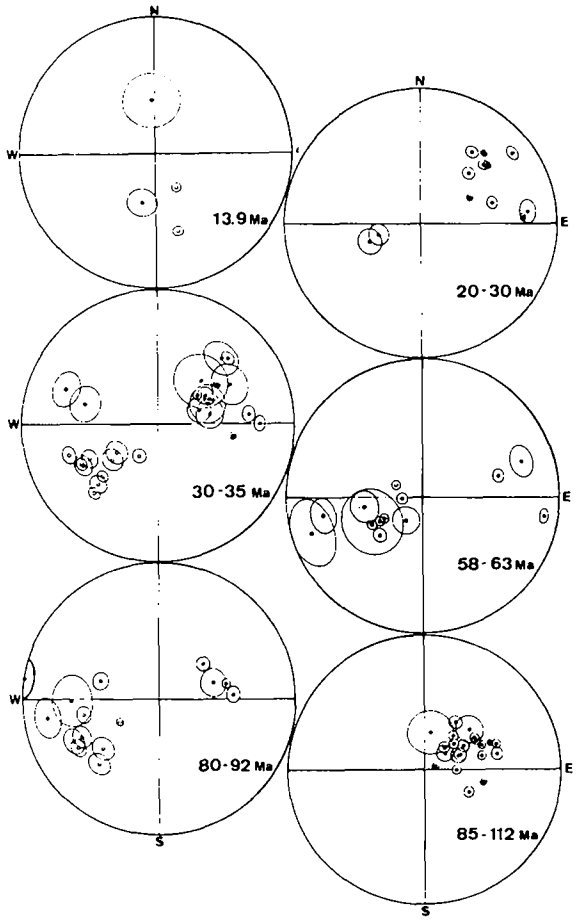


Fig. 3. Mean magnetic directions and 95% confidence circles of sites for three periods of 58–63 Ma, 80–92 Ma and 85–112 Ma together with those of younger periods of our previous papers [10,23]. Projections are equal area, solid (open) symbols on the lower (upper) hemisphere.

fission track method for the welded tuffs: the Shunan and Abu Groups overlie the Kanmon Group with Cenomanian to Turonian age [17]. The welded tuffs have been held at somewhat lower temperatures since their formation.

(2) Dominant magnetic carrier is magnetite, which is hardly produced in an oxidizing environment at low temperature. We accept the result after tilt correction was the more plausible paleomagnetic estimate because we collected samples with primary magnetic component from many sites in a wide area and the error due to the attitude of strata was averaged to zero.

All samples probably acquired their magnetization during Cretaceous normal polarity chron between 118 and 83 Ma [26].

3.2. 80–92 Ma (the Ochi Group)

Nine additional sites were measured. All the directions showed clockwise-deflected declinations. The reliability of this easterly declination is substantiated by the presence of both normal and reversed polarities. Samples from six sites in the lower part of the Ochi Group (the Ichiibara Formation) have reversed polarity. Since an uncertainty of the order of 10% is introduced for the fission-track method of dating, their magnetization was acquired possibly during the reversed interval between magnetic anomaly 33 and 34 which occurred just after the end of Cretaceous normal polarity chron.

3.3. 58–63 Ma (the Sakurae Group)

Eight new sites have been measured and their data have been combined to the previously reported data. The equal area plots show considerable scatter ($\alpha_{95} = 12.0°$). The wide dispersion may be attributed to the secular variation of the geomagnetic field rather than due to the secondary magnetic component acquired after tectonic deformation, because no reduction in scatter is observed when the tilt correction is applied (precision parameter increases from 10.9 to 11.9). While all the polarities for upper part of the formation (N5, N4 and N3) are reversed, the lower part (N2) shows a composite of normal and reversed polarities. This could suggest that the formation was formed during the interval over both normal and reversed polarity epochs. Our samples have adequate age ranges for averaging out the secular variation.

4. Discussion

4.1. Apparent polar wandering path over the last 100 million years

We have constructed an apparent polar wandering path (APWP) using the results of this study and those of younger rocks of our previous studies [10,23]. Six virtual geomagnetic poles (VGPs) have been estimated in the period between 100 and 14 Ma (Table 2). Fig. 4 shows the mean pole for each stage (13.9 Ma, 20–30 Ma, 30–35 Ma, 58–63 Ma, 80–92 Ma and 85–112 Ma) to-

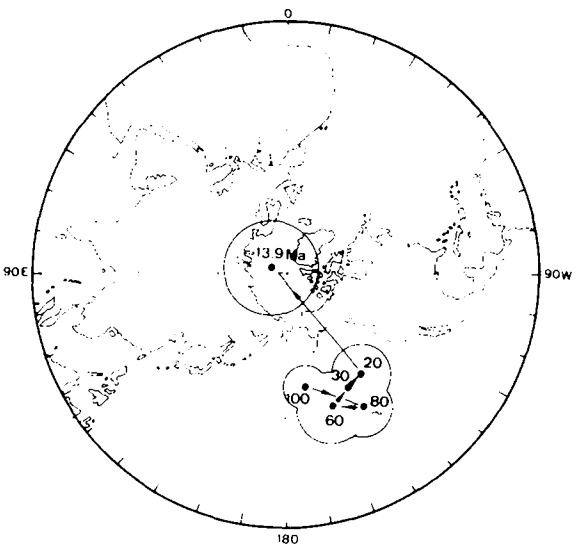


Fig. 4. An APWP with 95% confidence width of a late Mesozoic/Cenozoic period of Southwest Japan. The APWP consists of VGPs of six stages of 13.9 Ma, 20–30 Ma 30–35 Ma, 58–63 Ma, 80–92 Ma and 85–112 Ma. These stages are labelled by 13.9 Ma, 20 Ma, 30 Ma, 60 Ma, 80 Ma and 100 Ma, respectively. Polar stereographic projection.

gether with the 95% confidence interval.

The APWP for Southwest Japan is characterized by the succession of two different motions (see Fig. 4). Between 100 and 20 Ma, the VGPs are situated to the northwest of the Hawaiian Islands, and they underwent a small eastward displacement. From 20 to 10 Ma, the VGP abruptly moves to the present geographic north pole position. The amount of eastward drift between 100 and 20 Ma is 21°, while more than 50° is observed between 20 and 10 Ma.

The abrupt change of VGP between 20 and 10 Ma reflects the sudden clockwise rotation of Southwest Japan relative to the present geographic coordinate system proposed by Otofuji and Matsuda [10.23].

The gradual displacement between 100 and 20 Ma is also observed in the APWP from northern Eurasia [27] and western Europe [28]. The amount of the displacement of 15° for these areas is similar to that of Southwest Japan, although their VGPs move polewards. The eastward displacement of the APWP for Southwest Japan is ascribed to its clockwise tectonic rotation. Similarity between the APWPs prior to 20 Ma for Southwest Japan and the above two areas suggests that Southwest Japan and Eurasia occupied the same plate between 100 and 20 Ma.

4.2. Rotation versus age

The amount of rotation $R(t)$ of Southwest Japan with respect to Eurasia is calculated for each stage (t = 100, 80, 60, 30, 20 and 13.9 Ma) through a comparison of observed and expected declination values. We use Irving's [27] APWP of Eurasia to compute the expected geomagnetic field

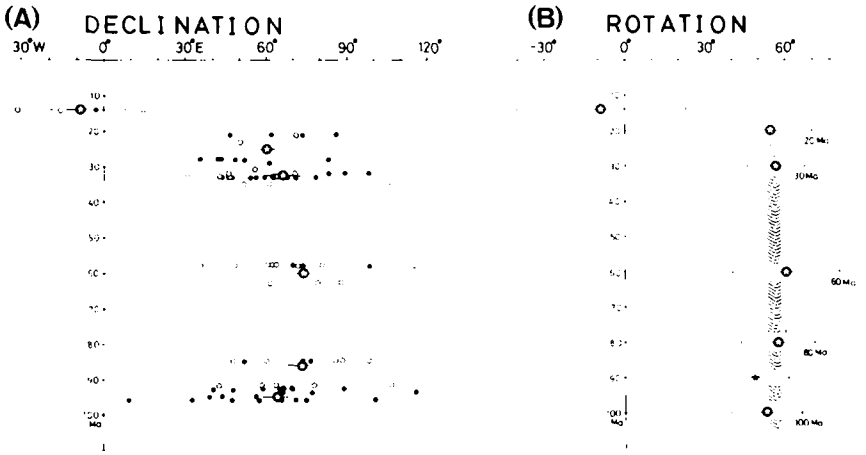


Fig. 5. A. Mean declination data versus age for each site and each stage of 13.9 Ma, 20–30 Ma, 30–35 Ma, 58–63 Ma, 80–92 Ma and 85–112 Ma. Closed (open) circles show the data with normal (reversed) polarity. B. Rotation with respect to Eurasia versus age for each stage of 13.9 Ma, 20 Ma, 30 Ma, 60 Ma, 80 Ma and 100 Ma. Rotation with respect to the eastern part of Eurasia is shown as a closed star. The error bar in declination and rotation for each stage are ΔD and ΔR values in Table 2. Shaded declination zone is the standard error of rotation ΔR_{ex} (see text).

direction at the representative point of Southwest Japan (134° E, 35° N). The amount of rotation R and its uncertainty ΔR are given

$$R = D_{ex} - D_{ob}; \quad \Delta R = \sqrt{(\Delta D_{ob})^2 + (\Delta D_{ex})^2}$$

where D_{ob} and D_{ex} are the observed and expected declination, and ΔD_{ob} and ΔD_{ex} are their uncertainty at 95% confidence ($\Delta D = \sin^{-1}[\sin(\alpha_{95})/\cos I]$ with α_{95} the radius of the 95% confidence circle and I the inclination). The values R and ΔR for six stages are listed in Table 2 and shown in Fig. 5 as a function of age. Declination is also plotted in Fig. 5.

It is clearly seen that almost the entire clockwise rotation of Southwest Japan took place between 20 and 10 Ma. Through 100 Ma and 20 Ma, the values of $R(t)$ have been, as a first-order solution, interpreted to be constant. Little rotational motion occurred between 100 and 20 Ma.

The value $R(t)$ can be fitted by a second-order polynomial, showing that the rate of rotation $R(t)$ has a maximum value of 59.9° at 58 Ma. The rifting process may have already progressed more or less between 60 and 20 Ma.

4.3. Amount of rotation

The arithmetical mean (R_{ex}) of the amount of rotation $R(t)$ of five stages (100, 80, 60, 30 and 20 Ma) is 56°. The standard error ΔR_{ex} is 3°, where $\Delta R_{ex} = \sqrt{[R(t) - R_{ex}]^2/n}$. We conclude that the amount of rotation with respect to Eurasia is $56 \pm 3°$ around the local pivot (134° E, 35° N).

The amount of rotation relative to the eastern margin of Eurasia is estimated. The paleomagnetic field directions of 100 Ma and 100–80 Ma from Southwest Japan are compared with the expected geomagnetic field direction which has been computed using the Cretaceous pole (72.8° N, 166.0° W, $\alpha_{95} = 5.7°$), recently obtained for the eastern margin of Eurasia (Korea, China and Siberia) [29–32]. The computed, expected declination and its uncertainty are $D = 20.2°$ and $\Delta D = 7.6°$. The amounts of rotation are $44 \pm 15°$ and $49 \pm 13°$, respectively (Fig. 5).

The amount of rotation of Southwest Japan with respect to the whole of Eurasia is larger than that with respect to the eastern margin of Eurasia (Fig. 5). This fact implies that the eastern part of Eurasia as well as Southwest Japan moved relative to the main part of Eurasia. The relative motion of the eastern part of the Eurasian continent may be associated with the collision of the Indian subcontinent and Eurasia [33]. We can describe the motion of Southwest Japan as follows: while the eastern margin of Eurasia, including Southwest Japan, was rotated with respect to the main part of Eurasia during last 100 Ma, Southwest Japan rotated clockwise by more than 40° with respect to the eastern margin of Eurasia.

The amount of rotation in our results is larger than those of previous works. Yaskawa [34] and Sasajima [35] estimated the amount of rotation using compilation of Cretaceous data set from the entire area of Southwest Japan as of 1973 and 1980, respectively. They estimated the amount to be 24° and 28° with respect to the Korean Peninsula. This is less than 65% of our estimate of 44° or 49°. Their smaller values are partly due to the use of magnetically unstable rocks (granite, granodiorite, andesite) without careful thermal demagnetization and lack of tilting correction. Hayashida and Ito [36] estimated 45° for the amount of rotation with respect to Eurasia using the paleomagnetic results for sedimentary rocks from the eastern part of Southwest Japan, while Otofuji et al. [37] obtained 47° using a data set obtained from Southwest Japan since 1980. Both values are also smaller than ours (45° and 47° vs. 56°). It should be noted that critical data set, by which Otofuji et al. [37] estimate the amount of rotation, is mainly brought from the eastern part of Southwest Japan far from our study area. These smaller amounts of rotation estimated for Southwest Japan including its eastern part would suggest that the eastern part has in fact experienced rotation smaller than our study area.

4.4. Fan shape opening of the southwestern part of the Japan Sea

The rotation of Southwest Japan appears to have begun by opening of the Japan Sea, according to geological evidence, from the basin floor of the Japan Sea [37]. The amount of rotation of a moving block is the strongest key to indicate the mode of opening for back arc basin. Opening modes are classified into three groups; (1) fan shape opening (Ligurian Sea); (2) the Atlantic type of parallel opening with symmetric magnetic lineations (Parece Vela, Shikoku, Bismarck, Lau,

Havre Basins and Fiji Plateau), and (3) pull-apart basin (opening related to leaky transform fault) (Gulf of California and Andaman Sea). Only the fan shape mode satisfies the paleomagnetic constraint of amount of rotation of more than 40° for Southwest Japan. It is hard to explain the large amount of rotation by the last two modes, because translation without rotation is required for block motion: less than 20° is claimed for clockwise rotation of Southwest Japan by the formation model of a pull-apart basin [15].

If Southwest Japan has been restored to the primary position prior to opening of the Japan Sea, a primary VGP of the Cretaceous period (100 Ma or 80–100 Ma) for Southwest Japan may be calculated [38] (Fig. 6). Southwest Japan is assumed to have been rotated through 42° or 47° about a rotation pole of 129°E, 34°N associated with the fan shape opening (Fig. 7). The primary VGP agrees with the Cretaceous VGPs for the eastern margin of Eurasia, strongly supporting our model of the fan shape opening.

Certain tectonic evidence is compatible with the view that Southwest Japan has been separated and rotated by the southwestern part of the Japan Sea, an opening which widens northeastwards; (1) the Hokusatsu bending, and (2) the width of the Japan Sea. First, several geological terranes (the Chichibu Group (Permian to lower Jurassic), Sambosan Group (Triassic to Jurassic), and Shimanto Group (upper Jurassic to lower Miocene)) are zonally arranged from north to south in the outer zone of Southwest Japan. This zonal structure is bent at the point of the Hokusatsu area (see Fig. 7) from ENE-SSW by about 40–60° [39]. The amount of this bending is quite consistent with that of the rotation of Southwest Japan inferred from the paleomagnetic viewpoint. This bending furthermore suggests that the rotation pivot existed near the western edge of Southwest Japan. Second, the width of the central part of the Japan Sea is about 600 km (see Fig. 7). Assuming the rotation pivot point of Southwest Japan at 129°E, 34°N, the expected width of the Japan Sea may be calculated. The amount of southeastward translation of eastern part of Southwest Japan is expressed as $s = L \times R$; where L is the distance between the pivot of rotation and the eastern part of Southwest Japan, and R is the angle of rotation in radians (45° = 0.8 rad). L is about 600 km, and therefore, translation is (600 × 0.8) km. This translation is comparable with the width of the Japan Sea.

An analogue of the proposed fan shape opening

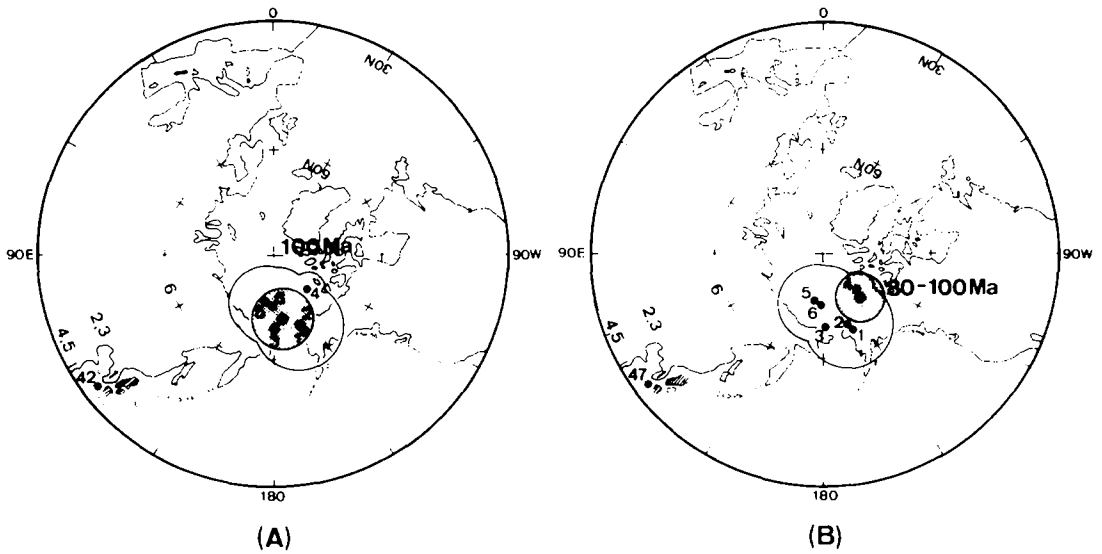


Fig. 6. Newly reconstructed VGPs with 95% confidence circle of Cretaceous period (100 Ma (A) and 80–100 Ma (B)) of Southwest Japan (shaded). Southwest Japan is rotated through 42° and 47° about the pivot of 129°E, 34°N, respectively. For comparisons are shown (with 95% confidence circles) the Cretaceous VGPs for *1*: Korean Peninsula [30]; *2*, *3*: North China (Upper and Lower Cretaceous) [31], *4*: South China (Lower Cretaceous) [31], *5*: Nanjin [32]; *6*: Siberia [29].

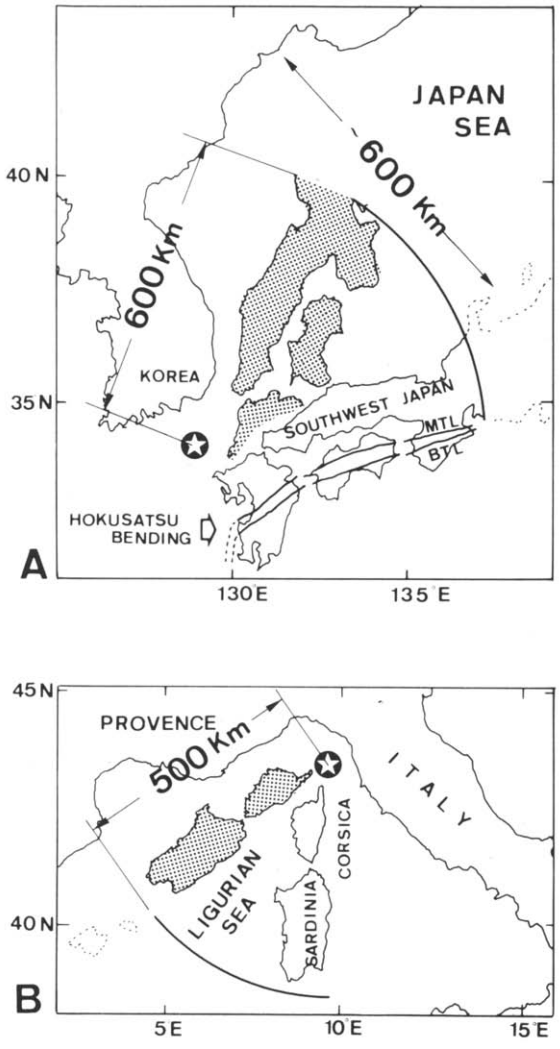


Fig. 7. A comparison of rotation of the continental sliver. A. Southwest Japan is rotated about the rotation pivot 129° E. 34° N through 47° associated with the fan shape opening of the southwestern part of the Japan Sea. *MTL*: Median Tectonic Line. *BTL*: Butsuzo Tectonic Line which is a boundary between the Sambosan and Shimanto Groups. B. Corso-Sardinia block is estimated to be rotated about the pivot 9°38′E. 43°22′N through 49° associated with the fan shape opening of the Ligurian Sea [41].

for the southwestern part of the Japan Sea would be found in Ligurian Sea placed between South France and Corso-Sardinia block (Fig. 7). Since Argand's [40] paleomorphological reconstruction in which the Corso-Sardinian block lies adjacent to South France, numerous paleomagnetic studies of Corso-Sardinia block and magnetic survey of the Ligurian Sea have been carried out to examine a rotational model for the Corso-Sardinia block [41–45]. The following facts are comparable with the Southwest Japan–southwestern Japan Sea system: (1) displacements of the land blocks are associated with fan shape mode opening of the basin between the land block and its mother continent; (2) both Southwest Japan and Corso-Sardinia basements partly consist of Paleozoic metamorphic complexes older than Permian, indicating that these blocks are micro-continents; (3) the amount of rotation is estimated to be more than 30° for both micro-continents, although the direction of rotation is opposite. The analogy is strengthened by the following two tectonic features:

(a) The duration of the rotational motion is quite short. Paleomagnetic work together with radiometric dating suggests that the duration is at least 1.5 Ma for Sardinia [44] and less than 1 Ma for Southwest Japan [37].

(b) The length of the moving block is about 600 km for both micro-continents.

The rotational motions of both Southwest Japan and Corso-Sardinia blocks are characterized by rifting of the continental lithosphere. Thick and ductile lithosphere may prevent the continental rim from complete rifting from its mother continent; progression of the splitting process is hampered in ductile material, and some rift valleys never evolve into an oceanic basin because of the large thickness of the lithosphere. The fan shape mode for the opening of a back arc basin probably reflects the mechanical character of the continental lithosphere, e.g., thickness, ductility and thermal condition.

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